

Heiner Litz

Curriculum Vitae

August 2026

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Overview

I am an Associate Professor at the University of California Santa Cruz and hold the Kumar Malavalli Endowed Chair of Storage Systems Research. I serve as the Director of the Center for Research in Systems and Storage (CRSS). My research spans computer architecture, storage, and systems to design next-generation microprocessors for scale-out data centers. My current interests revolve around re-designing the processor frontend (Icache, BTB, branch prediction) and the memory system, enabling efficient execution of warehouse-scale computing workloads. I am building full-stack prototypes, including applications, firmware, FPGAs, and ASICs. I have contributed to the Linux kernel. I am currently advising 7 Ph.D. students, and my research is supported by grants of over \$10M from NSF, Samsung, Google, Facebook, Intel, Meta, Nutanix, and ARM.

Work Experience

2024 Visiting Professor at the Massachusetts Institute of Technology (MIT), Cambridge
2024- Kumar Malavalli Endowed Professor for Storage Systems Research, UCSC
2023- Associate Professor at the University of California, Santa Cruz
2023- Director of the Center for Research in Systems and Storage at UCSC, Santa Cruz
2022-2026 Consulting Performance Exploration Engineer, ARM
2018-2023 Assistant Professor at the University of California, Santa Cruz
2018-2020 Consulting Researcher at Google, Mountain View
2017 Visiting Faculty at Google, Mountain View
2015-2016 Lecturer and Postdoctoral Researcher, Stanford University, with Prof. C. Kozyrakis
2012-2015 Postdoctoral Researcher, Stanford University, with Prof. D. Cheriton
2008 Hardware Engineering Intern, SUN Microsystems
2006-2012 Research Assistant, University of Heidelberg and Mannheim University

Education

2015 Postdoc in Electrical Engineering, Stanford University
Advisor: David Cheriton and Christos Kozyrakis
2011 PhD. in Computer Engineering, Mannheim University
Advisor: Professor Ulrich Bruening
2010 Visiting Researcher, University of Valencia
Advisor: Professor Jose Duato
2006 Diplom Informatik (M.Sc), Mannheim University

Honors and Awards

- IEEE MICRO Top Pick 2023 Award
- Best Paper Award, MICRO'22
- Intel Outstanding Researcher Award '21
- NSF CAREER Award
- Google Faculty Award
- Facebook Faculty Award
- IEEE MICRO Top Pick 2019 Award
- Memorable Paper Award, NVMW'17
- Dissertation with honors, Summa Cum Laude

Publications

Refereed Conference Papers

1. Huang, G., **Litz, Heiner**, and Y. Xu (2026). PIPM: Partial and Incremental Page Migration for Multi-host CXL Disaggregated Shared Memory. In: *Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, pp.1445–1460.
2. Jamilan, S., S. Kumar, and **Litz, Heiner** (2026). RIFS: Run-Time Invariant Function Specialization. In: *Proceedings of the 35th ACM SIGPLAN International Conference on Compiler Construction*, pp.40–52.
3. Liu, Y., M. Xie, S. Shi, Y. Xu, **Heiner Litz**, and C. Qian (2025). Outback: Fast and Communication-efficient Index for Key-Value Store on Disaggregated Memory. In: *Proceedings of the 51st International Conference on Very Large Data Bases (VLDB)*.
4. Wintz, P. K., Y. Sonmez, P. Griffioen, M. Xu, S. Oh, **Litz, Heiner**, R. G. Sanfelice, and M. Arcak (2025). Sharc: Simulator for Hardware Architecture and Real-time Control.
5. Zhao, Y., S. Oh, M. Xu, and **Litz, Heiner** (2025). ATR: Out-of-Order Register Release Exploiting Atomic Regions. In: *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture®*, pp.705–718.
6. Zhu, K., Y. Zhao, Y. Gao, P. Braun, T. A. Khan, **Heiner Litz**, B. Kasikci, and S. Deng (2025). From Optimal to Practical: Efficient Micro-op Cache Replacement Policies for Data Center Applications. In: *Proceedings of the International Symposium on Computer Architecture (HPCA)*.
7. Chakraborty, J., M. Dorier, P. Carns, R. Ross, C. Maltzahn, and **Heiner Litz** (2024). Thallus: An RDMA-based Columnar Data Transport Protocol. In: *Proceedings of the 2nd Workshop on Hot Topics in System Infrastructure (HotInfra)*.
8. Oh, S., M. Xu, T. A. Khan, B. Kasikci, and **Litz, Heiner** (2024). UDP: Utility-Driven Fetch Directed Instruction Prefetching. In: *51st International Symposium on Computer Architecture (ISCA)*. ISCA.
9. Xie, M., C. Qian, and **Heiner Litz** (2024). En4S: Enabling SLOs in Serverless Storage Systems. In: *Proceedings of the Symposium on Cloud Computing (SoCC)*.
10. Zhang, Y., N. Sobotka, S. Park, S. Jamilan, T. A. Khan, B. Kasikci, G. A. Pokam, **Litz, Heiner**, and J. Devietti (2024). RPG 2: Robust Profile-Guided Runtime Prefetch Generation. In: *27th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'24)*.
11. Jaliminche, L. N., C. N. Chakrabortii, C. Choi, and **Litz, Heiner** (2023). Enabling Multi-tenancy on SSDs with Accurate IO Interference Modeling. In: *2023 ACM Symposium on Cloud Computing (SoCC'23)*.
12. Liu, Y., S. Shi, M. Xie, **Litz, Heiner**, and C. Qian (2023). Smash: Flexible, Fast, and Resource-efficient Placement and Lookup of Distributed Storage. *ACM on Measurement and Analysis of Computing Systems (SIGMETRICS)*.
13. Ni, Y., P. Mehra, E. Miller, and **Litz, Heiner** (2023). TMC: Near-Optimal Resource Allocation for Tiered-Memory Systems. In: *2023 ACM Symposium on Cloud Computing (SoCC'23)*.
14. Zhang, Y., T. A. Khan, G. Pokam, B. Kasikci, **Litz, Heiner**, and J. Devietti (2023). Online COde Layout OptimizationS via OCOLOS. *IEEE Micro Top Pick*.
15. Jamilan, S., T. A. Khan, G. Ayers, B. Kasikci, and **Heiner Litz** (2022). APT-GET: profile-guided *timely* software prefetching. In: *EuroSys '22: Seventeenth European Conference on Computer Systems (Eurosys'22)*.
16. Khan, T. A., M. Ugur, K. Nathella, D. Sunwoo, **Litz, Heiner**, D. Jiménez, and B. Kasikci (2022). Whisper: Profile-Guided Branch Misprediction Elimination for Data Center Applications. In: *55nd Annual International Symposium on Microarchitecture (MICRO'22)*.

17. **Litz, Heiner**, G. Ayers, and P. Ranganathan (2022). CRISP: Critical Slice Prefetching. In: *Twenty-Seventh International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'22)*.
18. Song, S., T. A. Khan, S. Mahdizadeh-Shahri, A. Sriraman, N. K. Soundararajan, S. Subramoney, D. A. Jiménez, **Heiner Litz**, and B. Kasikci (2022). Thermometer: profile-guided btb replacement for data center applications. In: *ISCA '22: The 49th Annual International Symposium on Computer Architecture (ISCA'22)*.
19. Yuxuan, Z., T. A. Khan, G. Pokam, B. Kasikci, **Litz, Heiner**, and J. Devietti (2022). OCOLOS: On-line COde Layout OptimizationS. In: *55nd Annual International Symposium on Microarchitecture (MICRO'22)*.
20. Chakrabortii, C. and **Litz, Heiner** (2021). Reducing write amplification in flash by death-time prediction of logical block addresses. In: *14th ACM International Conference on Systems and Storage (SYSTOR)*.
21. Kargar, S., **Litz, Heiner**, and F. Nawab (2021). Predict and Write: Using K-Means Clustering to Extend the Lifetime of NVM Storage. In: *37th IEEE International Conference on Data Engineering (ICDE)*.
22. Khan, T. A., D. Zhang, A. Sriraman, J. Devietti, G. Pokam, **Litz, Heiner**, and B. Kasikci (2021). Ripple: Profile-Guided Instruction Cache Replacement for Data Center Applications. In: *48th International Symposium on Computer Architecture (ISCA'21)*.
23. Ayers, G., **Litz, Heiner**, C. Kozyrakis, and P. Ranganathan (2020). Classifying Memory Access Patterns for Prefetching. In: *Twenty-Fifth International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'20)*.
24. Chakrabortii, C. and **Litz, Heiner** (2020a). Improving the accuracy, adaptability, and interpretability of SSD failure prediction models. In: *11th ACM Symposium on Cloud Computing (SoCC)*.
25. Chakrabortii, C. and **Litz, Heiner** (2020b). Learning I/O Access Patterns to Improve Prefetching in SSDs. In: *European Conference on Machine Learning (ECML-PKDD'20)*.
26. Khan, T. A., A. Sriraman, J. Devietti, G. Pokam, **Litz, Heiner**, and B. Kasikci (2020). I-SPY: Context-Driven Conditional Instruction Prefetching with Coalescing. In: *53nd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO'20)*.
27. Ayers, G., N. P. Nagendra, D. I. August, H. K. Cho, S. Kanev, C. Kozyrakis, T. Krishnamurthy, **Litz, Heiner**, T. Moseley, and P. Ranganathan (2019). AsmDB: understanding and mitigating front-end stalls in warehouse-scale computers. In: *46th International Symposium on Computer Architecture (ISCA'19)*.
28. Ni, Y., J. Zhao, **Litz, Heiner**, D. Bittman, and E. L. Miller (2019). SSP: Eliminating Redundant Writes in Failure-Atomic NVRAMs via Shadow Sub-Paging. In: *52nd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO'19)*.
29. Grossman, S., **Litz, Heiner**, and C. Kozyrakis (2018). Making pull-based graph processing performant. In: *23rd ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*.
30. Hashemi, M., K. Swersky, J. Smith, G. Ayers, **Litz, Heiner**, J. Chang, C. Kozyrakis, and P. Ranganathan (2018). Learning Memory Access Patterns. In: *35th International Conference on Machine Learning (ICML)*.
31. Klimovic, A., **Litz, Heiner**, and C. Kozyrakis (2018b). Selecta: Learning Heterogeneous Cloud Storage Configuration for Data Analytics. In: *USENIX Annual Technical Conference (ATC'18)*.
32. **Litz, Heiner**, A. Klimovic, and C. Kozyrakis (2017). ReFlex: Remote Flash == Local Flash. In: *22nd International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'17)*.
33. **Litz, Heiner**, B. Braun, and D. R. Cheriton (2016). EXCITE-VM: Extending the Virtual Memory System to Support Snapshot Isolation Transactions. In: *25th International Conference on Parallel Architectures and Compilation Techniques (PACT'25)*.

34. **Litz, Heiner**, D. Cheriton, A. Firoozshahian, O. Azizi, and J. P. Stevenson (2014). SI-TM: reducing transactional memory abort rates through snapshot isolation. In: *19th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'14)*.
35. Fröning, H., M. Nüssle, **Litz, Heiner**, C. Leber, and U. Brüning (2013). On achieving high message rates. In: *13th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid'13)*.
36. Leber, C., B. Geib, and **Litz, Heiner** (2011). High Frequency Trading Acceleration using FPGAs. In: *21st International Conference on Field Programmable Logic and Applications (FPL'21)*.
37. Fröning, H., M. Nüssle, **Litz, Heiner**, and U. Brüning (2010). A Case for FPGA based Accelerated Communication. In: *9th International Conference on Networking (ICN'9)*.
38. **Litz, Heiner**, M. Thürmer, and U. Brüning (2010). TCCluster: A Cluster Architecture Utilizing the Processor Host Interface as a Network Interconnect. In: *2010 IEEE International Conference on Cluster Computing (CLUSTER)*.
39. Fröning, H., **Litz, Heiner**, and U. Brüning (2009). Efficient Virtualization of High-Performance Network Interfaces. In: *8th International Conference on Networks (ICN'8)*.
40. **Litz, Heiner**, H. Fröning, M. Thürmer, and U. Brüning (2009). An FPGA based Verification Platform for HyperTransport 3.x. In: *19th International Conference on Field Programmable Logic and Applications (FPL'19)*.
41. **Litz, Heiner**, H. Fröning, M. Nüssle, and U. Brüning (2008). VELO: A Novel Communication Engine for Ultra-low Latency Message Transfers. In: *37th International Conference on Parallel Processing (ICPP'08)*.
42. Fröning, H., M. Nüssle, D. Slognat, **Litz, Heiner**, and U. Brüning (2006). The HTX-Board: A Rapid Prototyping Station. In: *3rd annual FPGAWorld Conference*.
43. Ding, Y. and **Litz, Heiner** (2005). Creating Multiplatform User Interfaces by Annotation and Adaption. In: *11th International Conference on Intelligent User Interfaces (IUI'11)*.
44. Ding, Y., **Litz, Heiner**, and D. Pfisterer (2004). A graphical single-authoring framework for building multi-platform user interfaces. In: *9th international conference on Intelligent user interface (IUI'9)*.

Journal Papers

1. **Litz, Heiner**, J. Gonzalez, A. Klimovic, and C. Kozyrakis (2021). RAIL: Predictable, Low Tail Latency for NVMe Flash. In: *ACM Transactions on Storage (TOS)*.
2. Nagendra, N. P., G. Ayers, D. I. August, H. K. Cho, S. Kanev, C. Kozyrakis, T. Krishnamurthy, **Litz, Heiner**, T. Moseley, and P. Ranganathan (2020). AsmDB: Understanding and Mitigating Front-End Stalls in Warehouse-Scale Computers. *IEEE Micro*.
3. **Litz, Heiner**, R. J. Dias, and D. Cheriton (2015). Efficient Correction of Anomalies in Snapshot Isolation Transactions. *ACM Transactions on Architecture and Code Optimization (TACO)*.

Refereed Workshop Papers

1. Klimovic, A., **Litz, Heiner**, and C. Kozyrakis (2018a). Learning Heterogeneous Cloud Storage Configuration for Data Analytics. In: *SysML Conference*.
2. **Litz, Heiner**, A. Klimovic, and C. Kozyrakis (2018). ReFlex: Remote Flash == Local Flash. In: *9th Annual Non-Volatile Memories Workshop (NVMW)*.
3. Wang, B., **Litz, Heiner**, and D. R. Cheriton (2014). HICAMP bitmap: space-efficient updatable bitmap index for in-memory databases. In: *10th International Workshop on Data Management on New Hardware (DAMON'10)*.
4. Chan, M., **Litz, Heiner**, and D. R. Cheriton (2013). Rethinking network stack design with memory snapshots. In: *14th USENIX Conference on Hot Topics in Operating Systems (HotOs'14)*.
5. **Litz, Heiner**, C. Leber, and B. Geib (Nov. 2011). DSL Programmable Engine for High Frequency Trading Acceleration. In: *4th Workshop on High Performance Computational Finance (WHPCF'11)*.

6. Fröning, H. and **Litz, Heiner** (2010). Efficient hardware support for the partitioned global address space. In: *24th IEEE International Symposium on Parallel & Distributed Processing, Workshops and Phd Forum (IPDPSW)*,
7. **Litz, Heiner**, H. Fröning, and U. Brüning (2010). HTAX: A Novel Framework for Flexible and High Performance Networks-on-Chip. In: *4th Workshop on Interconnection Network Architectures: On-Chip, Multi-Chip (INA-OCMC'4)*.
8. Kalisch, B., A. Giese, **Litz, Heiner**, and U. Brüning (2009). HyperTransport 3 Core: A Next Generation Host Interface with Extremely High Bandwidth. In: *1st International Workshop on HyperTransport Research and Applications (WHTRA'09)*.
9. **Litz, Heiner**, H. Fröning, and U. Brüning (2009). A HyperTransport 3 Physical Layer Interface for FPGAs. In: *6th International Symposium on Applied Reconfigurable Computing (ARC'6)*.
10. Nüssle, M., H. Fröning, A. Giese, **Litz, Heiner**, D. Slognat, and U. Brüning (2007). A Hypertransport based low-latency reconfigurable testbed for message-passing developments. In: *2nd Workshop Kommunikation in Clusterrechnern und Clusterverbundsystemen (KiCC 2007)*.

Teaching Experience

Instructor, CSE12: "Computer Systems and Assembly Language and Lab"	'24
Instructor, CSE120: "Computer Architecture"	'25,'24,'23,'22,'21,'20,'19,'18
Instructor, CSE125: "Logic Design with Verilog"	'22,'21,'20,'19
Instructor, CSE220: "Advanced Computer Architecture"	'25,'23,'22
Instructor, CSE226: "Advanced Parallel Computing"	'21,'20,'19

Grants, Gifts & Contracts

I have acquired as PI or Co-PI \$10.4 million in external research grants.

2026	H. Litz "Samsung Sponsored Research"	\$1,000,000
2023	H. Litz "Google gift"	\$90,000
2022	H. Litz "Google gift"	\$90,000
2022	R. Sanfelice, H. Litz, A. Halder "CPS Frontier NSF Large"	\$5,758,472
2022	H. Litz "Intel research contract"	\$150,000
2022	H. Litz "Samsung gift"	\$75,000
2021	H. Litz "Google gift"	\$60,000
2021	H. Litz "Facebook gift"	\$60,000
2020	H. Litz "NSF CAREER award"	\$529,995
2020	D. Long and H. Litz "NSF IUCRC Phase II grant"	\$500,000
2020	H. Litz "Samsung gift"	\$70,000
2020	H. Litz "Samsung research contract"	\$120,000
2020	R. Sanfelice, H. Litz, A. Halder "UCSC center-scale seed funding award"	\$75,000
2018	H. Litz and B. Paten "Accelerating Genomics Workloads with Smart SSDs" WD	\$960,000
2018	H. Litz "FoMR: Improving Microprocessor IPC for Data Center Workloads" NSF/Intel	\$360,000
2018	H. Litz and R. Felice "Optimized Trajectory planning for Automotive Vehicles", NXP	\$50,000
2017	H. Litz "NVMe Device Level support for Quality of Service". Samsung	\$100,000
2017	H. Litz "Data Center Flash Storage Disaggregation". Broadcom	\$100,000

CRSS Industrial Memberships

As Director of CRSS, I have acquired \$760,000 in industry funding.

2026	Cerabyte	\$15,000
2026	MARVEL	\$50,000
2026	Nutanix	\$50,000
2025	Cerabyte	\$15,000
2025	ARM	\$50,000
2025	MARVEL	\$50,000
2025	Nutanix	\$50,000
2024	Cerabyte	\$15,000
2024	Intel	\$50,000
2024	ARM	\$50,000
2024	MARVEL	\$50,000
2024	Nutanix	\$50,000
2023	Cerabyte	\$15,000
2023	Intel	\$50,000
2023	ARM	\$50,000
2023	MARVEL	\$50,000
2023	Nutanix	\$50,000
2023	Meta	\$50,000

Patents

Benjamin Geib, Heiner Litz, Mondrian Nuessle. "Circuit arrangement for connection interface". DE102011009518B4, EP2668763A2, US20140207881, filed 2011, assigned 2014.

Graduated Students

Ph.D. Students

2026 Surim Oh (Meta)
 2026 Saba Jamilan (PhD)
 2025 Lokesh Jaliminche (Aerospike)
 2025 Minghao Xie (Anyscale)
 2022 Yuanjian Ni (Meta)
 2021 Chandranil Chakrabortii (Trinity College)

MS Students

2025 Jayjeet Chakraborti (Marvel)
 2025 Mingsheng Xue (Hygon)
 2024 Peter Braun (ARM)
 2022 Karthik Rajesh Kain (Intel)
 2021 Maxwell Bradley (ScaleUpStream)
 2020 Sachet Mittal (Amazon AWS)

Current Students

Yinyuan Zhao (PhD)
 Chandrashish Mazumdar (PhD)
 Yuanpeng Liao (PhD)
 Gangqi Huang (PhD), co-advised with Yuanchao Xu
 Pete Wilcox (PhD)
 Moonseop Kim (PhD)
 Sun Tanaka (PhD)

Service

Chair

NVMW'24 - Program Chair
HOTCHIPS'23 - Program Chair
ASPLOS'23 - Sponsorship Co-Chair

Editor

IEEE MICRO - Associate Editor

Technical Program Comittee

ASPLOS'27
HPCA'27
ISCA'26
MICRO'26
ASPLOS'26
ISCA'25
HPCA'25
ASPLOS'25
ISCA'24
ASPLOS'24
MICRO'24
HPCA'24
ISCA'23
HPCA'23
ASPLOS'23
IISWC'22
MICRO'22
ISCA'22
HOTCHIPS'22
NSF'22 Panelist
NVMW'22
Eurosys'22
MICRO'21
ISCA'21
MICRO'20
Usenix ATC'21
Systor'20
Usenix ATC'20
SOCC'19
MICRO'19
HotOS'19
ISCA'19
HPCA-IS'19
PPoPP'19
SOCC'18
Micro'17
TACO'16
HPCA'15
ReCoSoC'11-16
HUCAA'13-15

Panelist

NSF'22 Panel DoE Basic Research Needs Panel NSF'19 Panel

Invited Talks**Storage Research**

2025 OCP
 2024 MARVELL
 2024 OCP
 2024 SNIA
 2023 HP
 2023 Kioxia
 2023 SNIA

Invited Talks**Datacenter Microprocessors**

2024 Intel
 2023 Intel
 September 2022 Texas A&M University
 May 2022 UC San Diego
 May 2022 Stanford University
 May 2022 UC Santa Barbara
 May 2022 UIUC
 May 2022 Industry Academia Partnership (IAP) Workshop
 April 2022 UT Austin
 April 2022 UC Berkeley
 March 2022 Google
 Jan 2022 Intel
 Nov 2021 ARM

Data center Operating Systems

October 2020 Intel
 August 2020 IBM
 August 2020 Nutanix
 August 2020 VMWare
 October 2019 Intel
 August 2019 Samsung
 August 2019 VMWare
 March 2019 Facebook
 December 2018 IAP Industry Academia Partnership
 August 2018 ITU University Copenhagen
 June 2018 Heidelberg University
 June 2018 SAP, Walldorf
 March 2017 UCSC
 Dec 2016 Facebook
 July 2016 Amazon AWS
 July 2016 CNEX
 July 2016 Google
 July 2016 Cavium
 Apr 2016 Ericsson
 Feb 2016 Broadcom
 Feb 2016 Stanford Platform Lab

Snapshot Isolation based Transactional Memory

Sep 2015 Samsung
July 2015 VMWare
July 2014 Heidelberg University
Jun 2013 HP Labs
Nov 2012 Stanford CIS

In Memory Deduplication

Nov 2013 SAP Labs
Oct 2013 High Performance Transaction Systems (HPTS)

High Frequency Trading with FPGAs

May 2012 Oracle Labs

HyperTransport Cache Coherent Interconnect and its Applications

Aug 2011 University of Valencia
Aug 2011 University of Albacete
Mar 2011 KLA-Tencor
Mar 2008 SUN Microsystems

Status

US and German Dual-Citizenship